

Magnify test cases

Every requirement of the Certified Invoicing System specification, the test that demonstrates it, and the result expected. Clause numbers refer to RRA's *Technical Specification of CIS for VSDC, Ver. 1.0* and the *VSDC API Specification v1.0.4*.

SYSTEM Magnify	VENDOR YoFinder Ltd	VERSION 1.1.50	COMPONENTS Manager · Till · Tablet
SDC ROUTE Local VSDC	DOCUMENT Draft 1		

KEY · 52 REQUIREMENTS IN TOTAL	
BUILT	Written and covered by automated test; nothing further is required from the Authority for it to work. 34 requirements
AWAITING CONTROLLER	Built as far as it can be, and blocked until the Authority supplies a controller, the RRA logo, an MRC, or watermark guidance. 16 requirements
ENVIRONMENTAL	A property of the hardware Magnify is installed on, satisfied by the equipment and its certification rather than by software. 2 requirements

There is no fourth status. Every requirement that could be met without something issued by the Authority has been met.

What the sixteen are waiting for. Section A and B-06 describe the conversation between Magnify and a Sales Data Controller — waiting for its response before printing, recognising that it is unavailable, ceasing operation when communication fails, transmitting the journal. These cannot be written against a controller that is not there, still less demonstrated.

D-02, D-03 and D-04 print values only a controller can supply: the SDC identifier and its clock, the internal data, the signature, and the QR code composed from them. E-01 to E-04 are the sale and refund receipts themselves, each of which carries that signature. D-05 and D-08 await artwork and an identifier issued by the Authority — the RRA logo and the MRC. E-05 additionally awaits the Authority's guidance on rendering a watermark on thermal paper, which no printer of that type overlays natively.

We have requested a test controller and will complete and verify all sixteen on receiving access. The rules they depend on are already written and tested — a settled sale cannot be cancelled, a refund references its original and is permitted once, receipt numbering is gap-free across concurrent tills — so what remains is connection rather than construction.

The status against each requirement is stated as it stands. Clause §7.10 requires the system to have no function beyond those its documentation states, so this document reports the position accurately rather than favourably. Where a requirement is incomplete it says so, and says why.

APPROACH

How the work was done, so that the statuses which follow can be weighed.

Checked against the Authority's own figures, not against our reading of the rules

The worked example in clause 13.1 is reproduced exactly. Ringing up the same bill — bread at 1,000 exempt, gouda at $33,600 \times 0.2$ less 25%, gum at 60×5 — Magnify prints:

TOTAL	6340.00
TOTAL A-EX	1000.00
TOTAL B-18.00%	5340.00
TOTAL TAX B	814.58

That last figure is the one worth naming. Taking 18% out of each line and adding the results gives 814.57. Taking it from the bracket total gives 814.58, which is what clause 13.1 prints. Magnify computes per-bracket tax from the bracket total for that reason.

The JSON sample in the VSDC API specification is likewise reproduced. It shows `prc 200000` with `taxAmt 30508`, which is 18% taken *out* of the price rather than added to it. Menu prices in Magnify are treated as tax-

inclusive on the same basis, so certification does not change what a guest is charged.

Where the specification says stop, the system stops

A journal that cannot be written stops the till, rather than allowing receipts nothing records (clause 7.14). A receipt left half-printed stops further sales, rather than leaving a sale registered without one (clause 7.15). A settled bill cannot be cancelled, edited or deleted; the only reversal is a refund referencing the original, permitted once (clause 7.17). There is no setting anywhere by which VAT can be turned off – the decision is made when the installer is built.

Covered by test

245 automated tests cover the fiscal behaviour and run on every change to the software: the tax arithmetic in every bracket and at rounding boundaries, receipt numbering under two terminals settling simultaneously, the message sent to the controller, the receipt layout, the daily reports against each of the twenty items in clause 18.1, recovery of an interrupted print, and the refusals above. They are part of a suite of 661 covering the system as a whole.

Reported as it stands

Where a requirement is incomplete this document says so and says why. Three contradictions between these documents were found and corrected before submission rather than left to be discovered.

A · CONNECTION TO THE SALES DATA CONTROLLER

The clauses that decide whether a receipt may exist at all. Everything else is downstream of these.

Every requirement in this section describes behaviour between Magnify and a Sales Data Controller. Waiting for its response before printing, recognising that it is unavailable, ceasing operation when communication fails – none of which we are able to implement or demonstrate without access to a controller.

The work that does not depend on a controller is complete: the sales message is built in the format the API specification defines and reproduces the worked example in that document, and the receipt carries the SDC information block, the internal data and signature grouped in fours, and the QR payload composed as clause 7.24.7 sets out. We will complete and verify this section on receiving access to a test instance, and would be glad to demonstrate the work already done at any stage.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
A-01	§10, §7.8	No receipt may print without a VSDC response.	Stop the local VSDC service. Settle a bill.	No receipt prints. The till states the VSDC is unavailable.	AWAITING CONTROLLER

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
A-02	§7.9	The system detects VSDC status independently and informs the user on error.	Stop the VSDC while the till is idle, without attempting a sale.	The till shows the fault on its own, before anyone tries to bill.	AWAITING CONTROLLER
A-03	§24.7.9	Operation stops on communication error or power interruption; printing resumes once cleared.	Kill the VSDC mid-service, then restart it.	Billing halts, then resumes without data loss or a skipped number.	AWAITING CONTROLLER
A-04	§7.15	A sale amount cannot be registered without simultaneously printing a receipt.	Attempt to close a bill with the printer offline.	The sale is not registered. No orphaned record is created.	AWAITING CONTROLLER
A-05	VSDC §2.2(4)	After 24 hours with no internet, the VSDC stops issuing receipt numbers.	Disconnect the internet, leaving the VSDC running. Bill normally, then exceed 24 hours.	Billing continues during the window; stops cleanly at expiry with a stated reason.	AWAITING CONTROLLER

B · NUMBERING AND THE JOURNAL

What RRA audits hardest. A repeated or skipped number is the defect these clauses exist to catch.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
B-01	§7.3	Consecutive numbering, one ascending sequence per receipt type, no gaps.	Issue 50 mixed receipts. Export the journal and check every sequence.	Each type runs 1..n with no gap and no repeat.	BUILT
B-02	§7.3	Two tills billing simultaneously cannot take the same number.	Settle concurrently from two terminals against one outlet, repeatedly.	Numbers are consecutive across both. No duplicate at any load.	BUILT
B-03	§7.24.4, §7.25	Both counters print as A/B RT — per type, and across all types.	Issue a sale after several refunds and copies. Read the printed line.	e.g. RECEIPT NUMBER: 168/258 NS , both advancing correctly.	BUILT
B-04	§7.3	A number claimed by an abandoned sale must not leave a gap.	Force a failure after the number is claimed but before the sale commits.	The number is released with the transaction. The next sale takes it.	BUILT
B-05	§7.14	An electronic journal records every receipt; the system will not operate without it.	Inspect the journal after a full trading day.	Every receipt present, in order, with its amounts and signature.	BUILT
B-06	§6.5	Complete journal data for NS and NR is sent to the VSDC in text form.	Issue sales and refunds, then inspect what reached the VSDC.	Journal text transmitted for every NS and NR receipt.	AWAITING CONTROLLER

C · TAX CALCULATION

Verified against RRA's own worked examples rather than against our reading of them — both the JSON sample in the API specification and the printed receipt in the CIS specification.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
C-01	§4.1	Tax rates labelled A, B, C, D at indexes 1–4; B is 18%, A is exempt.	Sell one item in each bracket on a single bill.	Each line carries its label. Only B attracts tax.	BUILT
C-02	API sample	Tax is extracted from a VAT-inclusive price, not added to it.	Sell one item at 200,000.	taxblAmt 200000 , taxAmt 30508 . Total charged stays 200,000.	BUILT
C-03	§13.1	The specification's own sample bill reproduces exactly.	Ring up bread 1,000 exempt; gouda 33,600 × 0.2 less 25%; gum 60 × 5.	TOTAL 6340.00 · A-EX 1000.00 · B 5340.00 · TAX B 814.58	BUILT
C-04	§13.1	Per-bracket tax is computed from the bracket total, not by summing rounded lines.	Bill two standard-rated lines whose individual taxes round down.	814.58 from the bracket, not 814.57 from the lines.	BUILT
C-05	§7.5	Amounts display and print to two decimal places.	Inspect the till display and a printed receipt.	Two decimals throughout, dot separator.	BUILT
C-06	§7.22, §7.23	Every programmed rate above zero prints on every receipt, used or not. A zero rate prints only when used.	Sell a single exempt item. Then a single standard-rated item.	B appears on both. A appears only on the bill that used it.	BUILT
C-07	§13	Discounts reduce taxable amount and tax proportionally, and are declared per line.	Apply 25% to one line of a mixed bill.	dcRt 25 , dcAmt and charge reconcile to the supply amount exactly.	BUILT

D · RECEIPT CONTENT

The printed article. Clause §7.24 fixes the SDC block precisely, including how the QR code is composed.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
D-01	§13.1	Header carries trade name, address and TIN.	Print any receipt.	All three present above the item list.	BUILT
D-02	§7.24.1–.3	SDC block: the designation, VSDC date and time as dd/mm/yyyy hh:mm:ss , and SDC ID .	Print a normal sale.	Block present, times taken from the VSDC and not the till.	AWAITING CONTROLLER
D-03	§7.24.5–.6	Internal data and signature printed in groups of four separated by dashes.	Print a normal sale and inspect both strings.	TE68-SLA2-34J5-EAV3-... — dash after every fourth character.	AWAITING CONTROLLER
D-04	§7.24.7	QR code composed exactly as specified.	Scan the printed QR and compare against the receipt fields.	Decodes to the format shown below the table.	AWAITING CONTROLLER

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
D-05	§7.29	The official RRA logo prints on every receipt, of every type.	Print one of each receipt type.	Logo present on all five.	AWAITING CONTROLLER
D-06	§7.27	An item counter shows the number of items on the receipt, excluding voids.	Bill three items, void one, print.	ITEMS NUMBER 2 .	BUILT
D-07	§7.28	The last line continues or reprints after power failure or paper-out recovery.	Remove paper mid-print, reload. Repeat with power.	Receipt completes without loss or duplication.	BUILT
D-08	§7.4, §7.7	MRC and a unique software version print, and the version is verifiable by RRA staff.	Print a receipt; check the version shown in the application.	MRC on the receipt; version reads 1.1.50 and matches the build.	AWAITING CONTROLLER

QR payload, per §7.24.7

```
invoice_date(ddmmyyyy) # time(hhmmss) # sdc_number # sdc_receipt_number
# internal_data # receipt_signature
```

E · RECEIPT TYPES

Five types, and strict rules about which may be issued when. The refund rules are the operationally surprising ones.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
E-01	§13	Normal sale (NS) is the ordinary receipt offered to the client.	Settle a bill.	Receipt labelled NS, signed, handed over.	AWAITING CONTROLLER
E-02	§7.17	A transaction cannot be corrected without cancelling the original, which must be referenced.	Refund a settled bill.	Refund receipt prints REF. NORMAL RECEIPT# naming the original.	AWAITING CONTROLLER
E-03	§7.17	An original may be cancelled only once.	Refund a bill, then attempt to refund it again.	The second attempt is refused, and says why.	AWAITING CONTROLLER
E-04	§13.2	Refund amounts print negative throughout — lines, totals, tax and tender.	Refund a mixed-bracket bill.	Every figure negative, including per-bracket tax.	AWAITING CONTROLLER
E-05	§7.18, §11	Only one original prints. A reprint is a COPY, watermarked, with the disclaimer.	Reprint an issued receipt.	COPY above the items and as a watermark; THIS IS NOT AN OFFICIAL RECEIPT below totals, at least twice the size of the amount text.	AWAITING CONTROLLER

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
E-06	§11	Training and proforma tickets carry the same designation and disclaimer.	Issue one of each.	TRAINING / PROFORMA marked as above. Neither is signed.	BUILT
E-07	§6.3.6	Training and proforma receipts are not digitally signed.	Inspect a training ticket.	No signature block. Not counted as a sale.	BUILT
E-08	§12	All corrections are made before the receipt is approved.	Amend a bill before settling, then settle.	Amendments possible before approval, impossible after.	BUILT

F · REPORTS

Clause §18 lists twenty items the daily report must contain. Each is a separate line of evidence.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
F-01	§7.6	Z daily report — the full day, 00:00:00 to 23:59:59.	Trade a full day, then produce the Z report.	Report covers the whole day and closes it.	BUILT
F-02	§7.6	X daily report — from the last Z report to now.	Produce X mid-service, twice.	Second X includes everything since the last Z, not since the last X.	BUILT
F-03	§18.1.5–.9	NS totals and count, NR totals and count, all tax-inclusive.	Trade a day containing sales and refunds. Read the report.	Four figures present and reconciling to the journal.	BUILT
F-04	§18.1.10–.11	Taxable and tax amounts per rate, split between NS and NR.	As above, with mixed brackets.	Per-bracket split shown separately for sales and refunds.	BUILT
F-05	§18.1.14–.16	Counts and amounts for copies, training and proforma receipts.	Issue several of each, then report.	Each type counted separately with its total.	BUILT
F-06	§18.1.17–.18	Sales split by means of payment, and all discounts.	Trade using cash, card, mobile money and credit, with discounts.	Each tender totalled separately; discounts totalled.	BUILT
F-07	§18.1.12, .20	Opening deposit and number of incomplete sales.	Open with a float; abandon a bill; report.	Both figures present.	BUILT
F-08	§7.26	An audit interface exposing software settings and the database.	Demonstrate the interface to an auditor.	Settings and records viewable without developer tools.	BUILT

G · STOCK AND OPERATIONS

Clause §7.30 carries an exemption that matters for a restaurant: prepared dishes registered as service items are not blocked by stock.

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
G-01	§7.11	Inventory control with input, removal, and independent status reports.	Receive stock, consume it, report.	Movements recorded; status report produced independently of sales.	BUILT

ID	CLAUSE	REQUIREMENT	TEST	EXPECTED	STATUS
G-02	§7.30	No receipt for goods when stock is short. Service items are exempt.	Attempt to sell a stocked good below its quantity, then a service item.	Goods refused; service item permitted.	BUILT
G-03	§7.31	Closing stock for any date chosen by the user.	Request closing stock for a past date.	Figures returned for that date, not today.	BUILT
G-04	§7.12	Cash deposits and withdrawals can be registered.	Record an opening float and a mid-service withdrawal.	Both recorded and reflected in the daily report.	BUILT
G-05	§7.13	A payment can be registered using different means.	Settle using each supported tender.	Each maps to its RRA payment code and prints correctly.	BUILT
G-06	§7.16	An amount cannot be registered without identifying the good or service.	Attempt to ring up a bare amount with no item.	Impossible — every line references a catalogue item.	BUILT
G-07	API §4.17	Every item carries a unique item code.	Export the catalogue and check for duplicates.	No code appears twice, including the same product sold at two stations.	BUILT
G-08	§7.2	TIN is reprogrammable only in service mode, and only via a reset that deletes all data held under the previous TIN.	Attempt to change the TIN in normal operation, then via service mode.	Refused in normal operation; permitted only with a full reset.	BUILT
G-09	§9	Functionality is unaffected by other equipment on the same host.	Run alongside other software and peripherals on the till.	No interruption to billing or printing.	ENVIRONMENTAL
G-10	§8	Host device approved by RURA or compliant with international safety standards.	Supply the till hardware's certification.	Documentation provided for the deployed hardware.	ENVIRONMENTAL

H · CERTIFICATION DOCUMENT PACK

The administrative documents RRA requires with a certification application. The list is identical for OSDC and VSDC, so the paperwork does not decide which route is taken.

#	DOCUMENT	HELD BY	NOTES	STATUS
1	Business Registration Certificate	YoFinder Ltd	RDB registration.	OBTAINED
2	Valid RSSB Clearance Certificate	YoFinder Ltd	Obtained 28 August 2026.	OBTAINED
3	Valid RRA Tax Clearance Certificate	YoFinder Ltd	—	OBTAINED
4	Physical address	YoFinder Ltd	—	OBTAINED
5	Software Documentation	YoFinder Ltd	Version 1. Sections 1–5 and 7–10 describe the software as built; section 6 states what is implemented and what awaits a controller. Revised for every change, as §7.10 requires.	DRAFTED

#	DOCUMENT	HELD BY	NOTES	STATUS
6	Software Support SLA	YoFinder Ltd	Complete. With a legal adviser for review before issue.	DRAFTED
7	Template (for suppliers with User)	YoFinder Ltd	Complete, as the Software Support Agreement signed with each user. With a legal adviser for review before issue.	DRAFTED
8	Software Support SLA Template (with RRA)	YoFinder Ltd	Complete. With a legal adviser for review before signature.	DRAFTED
9	Test Cases Document	This document	Draft. Statuses re-verified before submission.	DRAFT

All nine documents exist. Items 6, 7 and 8 are with a legal adviser, since they bind YoFinder Ltd to its users and to the Authority. Items 5 and 9 are revised as the software changes, which §7.10 requires of the documentation and which this document's own purpose requires of it.

Draft 6 · Magnify 1.1.50 · YoFinder Ltd · compiled 28 August 2026

Clause references: RRA Technical Specification of CIS for VSDC Ver. 1.0 (March 2018); RRA VSDC API Specification v1.0.4 (2022).

Status reflects the build at time of writing and must be re-verified before submission.